Lecture 10. Hypothesis Testing (Part 2)

Trend Tests, Stratified Tests, Matched Pairs, Renyi-Type Tests

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1 Review: The K-Sample Test

Right-censored data $\{(T_{ik}, \delta_{ik})\}$; pooled death times $t_1 < \cdots < t_D$; d_j, d_{jk} deaths and Y_j, Y_{jk} at risk at t_j (pooled / group k). With a common weight W,

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\}, \qquad \chi^2 = (Z_1, \ldots, Z_{K-1})\,\hat{\Sigma}_{K-1}^{-1}\,(Z_1, \ldots, Z_{K-1})^T \overset{H_0}{\approx} \chi^2_{K-1},$$

with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1} d_j, \qquad \hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1} d_j.$$

This omnibus test is **not sensitive to trends**.

2 Tests for Trend

2.1 Order-specific alternatives

$$H_0 : h_1(t) = \cdots = h_K(t) \;\; \forall t \le \tau \qquad \text{vs.} \qquad H_A : h_1(t) \le h_2(t) \le \cdots \le h_K(t) \;\; \forall t \le \tau,$$

with at least one strict inequality ($S_1 \ge \cdots \ge S_K$).

2.2 Trend test statistic

Objective: more power against the ordering.

Definition 10.1: Trend test

For scores $a_1 < a_2 < \cdots < a_K$,

$$Z = \frac{\sum_{k=1}^{K} a_k Z_k(\tau)}{\left(\sum_{k=1}^{K}\sum_{k'=1}^{K} a_k a_{k'} \hat{\sigma}_{kk'}\right)^{1/2}} = \frac{\mathbf{a}'\mathbf{Z}}{\sqrt{\mathbf{a}'\hat{\Sigma}\mathbf{a}}},$$

with the full $K \times K$ $\hat{\Sigma}$. Under H_0, $Z \approx N(0,1)$ (a combination of mean-zero martingales); reject if Z exceeds the upper α quantile (one-sided).

- **Choice of a_k**: under H_A, $\mathrm{E}\{Z_1(\tau)\} \leq \cdots \leq \mathrm{E}\{Z_K(\tau)\}$ (higher hazard, more observed than expected deaths), so the standard $a_k = k$ weights the high-expectation groups; other choices are possible.
- $\hat{\Sigma}$ is singular, but $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ for non-constant $\mathbf{a}$ (Exercise 10.1).

2.3 Invariance

Theorem 10.2: Invariance under linear transformations of the scores

Let $Z_{\mathbf{a}} = \mathbf{a}'\mathbf{Z}/\sqrt{V[\mathbf{a}'\mathbf{Z}]}$ and $\mathbf{b} = c\mathbf{a} + d\mathbf{1}_K$ with constants $c > 0$, d. Then $Z_{\mathbf{b}}(\tau) = Z_{\mathbf{a}}(\tau)$.

Proof [Supplement]. Since $\mathbf{1}'\mathbf{Z} = 0$ (so $\hat{\Sigma}\mathbf{1} = \mathbf{0}$),

$$\mathbf{b}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z} + d\,\mathbf{1}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z}, \qquad V[\mathbf{b}'\mathbf{Z}] = V[c\,\mathbf{a}'\mathbf{Z}] = c^2 V[\mathbf{a}'\mathbf{Z}].$$

Hence $Z_{\mathbf{b}} = Z_{\mathbf{a}}$ (the sign flips if $c < 0$). ■

E.g. $(1,2,3,4)$, $(0,1,2,3)$, $(-3,-1,1,3)$ give the same test.

Example 10.3: larynx cancer by stage

90 patients in stages I–IV; the KM curves are ordered (K&M Fig. 7.3). With $a_k = k$ and log-rank weights,

$$\mathbf{Z}(10.7) = (-7.5660,\ -3.0117,\ 2.9155,\ 7.6623),$$

$$\hat{\Sigma} = \begin{pmatrix} 12.0740 & -4.4516 & -6.2465 & -1.3759 \\ -4.4516 & 7.8730 & -2.7599 & -0.6614 \\ -6.2465 & -2.7599 & 9.9302 & -0.9238 \\ -1.3759 & -0.6614 & -0.9238 & 2.9612 \end{pmatrix}.$$

$\mathbf{a}'\mathbf{Z} = 25.806$ and $\mathbf{a}'\hat{\Sigma}\mathbf{a} = 48.152$, so

$$Z = \frac{25.806}{\sqrt{48.152}} = 3.72, \qquad Z^2 = 13.83, \qquad p < 0.0001.$$

(K&M report $Z^2 = 13.83$.) Tarone–Ware $Z = 4.06$, Gehan 4.22, Peto–Peto 4.13; all $p < 0.0001$: the hazards are in the expected order.

Remark 10.4

Use a trend test only with strong prior knowledge of the ordering.

3 Stratified Tests

3.1 Rationale

Comparisons often need covariate adjustment, e.g. 3 vs. 6 months of chemotherapy (IBCSG) adjusting for stage. Regression adjustment is sensitive to misspecification; **stratification** is a conservative alternative.

3.2 Set-up

Definition 10.5: Stratified K-sample test

$H_0 : h_{1s}(t) = h_{2s}(t) = \cdots = h_{Ks}(t)$, $t \in [0, \tau]$, for each stratum $s = 1, \ldots, M$.

1. Within stratum s, compute $Z_{ks}(\tau)$ and $\hat{\Sigma}_s$.
2. Sum: $Z_{k\cdot}(\tau) = \sum_s Z_{ks}(\tau)$, $\hat{\Sigma}_\cdot = \sum_s \hat{\Sigma}_s$ (independent strata).
3. $\chi^2 = [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]^T \overset{H_0}{\sim} \chi^2_{K-1}$.

Observed and expected deaths are compared within strata, so baseline differences across strata cannot masquerade as group effects.

Example 10.6: bone marrow transplant, stratified by MTX

The three disease groups (Lecture 9), now stratified by methotrexate prophylaxis (MTX) or not (NOMTX); log-rank ($W = 1$):

$$\text{MTX:} \quad (Z_1, Z_2, Z_3) = (1.2816,\ -2.4929,\ 1.2113),$$
$$\hat{\Sigma}_{MTX} = \begin{pmatrix} 6.23336 & -3.42915 & -2.80422 \\ -3.42915 & 5.60732 & -2.17818 \\ -2.80422 & -2.17818 & 4.98239 \end{pmatrix},$$
$$\text{NOMTX:} \quad (Z_1, Z_2, Z_3) = (-0.504,\ -11.235,\ 11.739),$$
$$\hat{\Sigma}_{NOMTX} = \begin{pmatrix} 9.6771 & -6.5535 & -3.1236 \\ -6.5535 & 13.9816 & -7.4281 \\ -3.1236 & -7.4281 & 10.5517 \end{pmatrix}.$$

Summing: $Z_{1\cdot} = 0.7776$, $Z_{2\cdot} = -13.7279$, $Z_{3\cdot} = 12.9503$, and

$$\hat{\Sigma}_\cdot = \begin{pmatrix} 15.91046 & -9.98265 & -5.92782 \\ -9.98265 & 19.58892 & -9.60628 \\ -5.92782 & -9.60628 & 15.53409 \end{pmatrix}.$$

With the first two:

$$\chi^2 = (0.7776,\ -13.7279) \begin{pmatrix} 15.9105 & -9.9827 \\ -9.9827 & 19.5889 \end{pmatrix}^{-1} \begin{pmatrix} 0.7776 \\ -13.7279 \end{pmatrix} = 13.19 \quad (2 \text{ df},\ p \approx 0.001).$$

Remark 10.7: Remarks

Good power when effects point the *same direction* in every stratum; otherwise they can cancel, and separate tests per stratum are preferable.

4 Matched Pairs

Each matched pair is a stratum: times (T_{1i}, T_{2i}), indicators $(\delta_{1i}, \delta_{2i})$, $H_0 : h_{1i}(t) = h_{2i}(t)$. With $K = 2$, per pair

$$Z_{1i}(\tau) = \sum_j W(t_j)\Big\{d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big\}, \qquad \sigma_{11i} = \sum_j W(t_j)^2 \frac{Y_{j1}}{Y_j}\Big(1 - \frac{Y_{j1}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1} d_j.$$

Derivation: within a single pair

Until the first time in the pair, $Y_j = 2$, $Y_{j1} = 1$.

1. $T_{1i} = T_{2i}$: $d_j = \delta_{1i} + \delta_{2i}$,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i} + \delta_{2i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i} - \delta_{2i}}{2},$$

$$\sigma_{11i} = W^2(T_{1i})\tfrac{1}{2}(1 - \tfrac{1}{2})\frac{2 - \delta_{1i} - \delta_{2i}}{2 - 1}(\delta_{1i} + \delta_{2i})$$

$$= \frac{W^2(T_{1i})}{4}(2 - \delta_{1i} - \delta_{2i})(\delta_{1i} + \delta_{2i}).$$

2. $T_{1i} < T_{2i}$: only the first time counts,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i}}{2}, \qquad \sigma_{11i} = \frac{W^2(T_{1i})}{4}(2 - \delta_{1i})\delta_{1i}.$$

3. $T_{1i} > T_{2i}$: $Z_{1i} = -W(T_{2i})\dfrac{\delta_{2i}}{2}$, $\quad \sigma_{11i} = \dfrac{W^2(T_{2i})}{4}(2 - \delta_{2i})\delta_{2i}$.

(Afterwards $Y_{j1}/Y_j \in \{0, 1\}$ and terms vanish.) With $(2 - \delta)\delta = \delta$:

$$Z_{1i} = \begin{cases} W(T_{1i})/2, & T_{1i} < T_{2i}, \delta_{1i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{1i} = 1, \delta_{2i} = 0, \\ -W(T_{2i})/2, & T_{2i} < T_{1i}, \delta_{2i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{2i} = 1, \delta_{1i} = 0, \\ 0, & \text{otherwise;} \end{cases}$$

$$\hat{\sigma}_{11i} = \begin{cases} W(T_{1i})^2/4, & \text{first case,} \\ W(T_{2i})^2/4, & \text{second case,} \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 10.8: Paired log-rank (sign-type) test

For constant W (e.g. log-rank), with D_1 (D_2) the pairs in which member 1 (2) has the event first,

$$Z_{1\cdot} = \frac{W}{2}(D_1 - D_2), \qquad \hat{\sigma}_{11\cdot} = \sum_{i=1}^{M} \hat{\sigma}_{11i} = \frac{W^2}{4}(D_1 + D_2), \qquad \frac{Z_{1\cdot}}{\sqrt{\hat{\sigma}_{11\cdot}}} = \frac{D_1 - D_2}{\sqrt{D_1 + D_2}} \overset{H_0}{\approx} N(0, 1).$$

A sign (McNemar-type) test: informative pairs go either way with probability $\frac{1}{2}$ under H_0 (Exercise 10.2).

Example 10.9: 6-MP vs. placebo matched pairs

$D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$: $Z = 15/\sqrt{21} = 3.27$, $p = 0.001$.

5 Renyi-Type Tests

5.1 Rationale

If the hazards **cross**, early and late differences cancel and the usual tests lose power. Renyi tests use the *supremum of absolute cumulative differences*, a censored-data Kolmogorov–Smirnov statistic.

5.2 Set-up and statistic

$H_0 : h_1 = h_2$ on $t \le \tau$ vs. $H_A : h_1(t) \neq h_2(t)$ for some t; notation as before with $K = 2$.

Definition 10.10: Renyi statistic

Cumulative weighted difference:

$$Z(t_i) = \sum_{t_j \le t_i} W(t_j)\Big[d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big], \qquad i = 1, \ldots, D,$$

with (τ the largest t_j with $Y_{j1}, Y_{j2} > 0$)

$$\sigma^2(\tau) = \sum_{t_j \le \tau} W^2(t_j)\frac{Y_{j1}}{Y_j}\frac{Y_{j2}}{Y_j}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

and two-sided statistic

$$Q = \frac{\sup\{|Z(t)|,\ t \le \tau\}}{\sigma(\tau)}.$$

Theorem 10.11: Null distribution

Under H_0, $Q \approx \sup\{|B(x)|,\ 0 \le x \le 1\}$, B standard Brownian motion (K&M Table C.5).

Remark 10.12: Why Brownian motion

$Z(t)$ is a martingale with variance $\sigma^2(t)$; in time $x = \sigma^2(t)/\sigma^2(\tau)$, $Z/\sigma(\tau)$ behaves like $B(x)$, and the supremum catches excursions even if $Z(\tau) \approx 0$.

Example 10.13: gastric cancer: chemotherapy vs. chemotherapy + radiotherapy

KM curves cross (K&M Fig. 7.5): chemo + radiation worse early, better later. Log-rank Renyi: $|Z(t_j)|$ peaks at $t_j = 315$ with 9.8 (K&M Fig. 7.4); $\sigma(2363) = 4.46$:

$$Q = 9.8/4.46 = 2.2, \qquad 0.05 < p < 0.06 \text{ (K\&M Table C.5)},$$

nearly significant, whereas the log-rank statistic $-2.15/4.46 = -0.482$ gives $p = 0.63$.

6 Selected exercises

Motivation. When the trend variance $\mathbf{a}'\hat{\Sigma}\mathbf{a}$ is positive, and why the paired log-rank test is an exact sign test.

Exercise 10.1 (When is the trend test defined?)

Write $\pi_{jk} = Y_{jk}/Y_j$ and $w_j = W(t_j)^2 d_j(Y_j - d_j)/(Y_j - 1)$, with $w_j = 0$ if $Y_j = 1$.
(a) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} = \sum_j w_j \Big\{ \sum_k \pi_{jk} a_k^2 - \Big(\sum_k \pi_{jk} a_k \Big)^2 \Big\} \geq 0$.
(b) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ if and only if at some death time with $w_j > 0$ the scores a_k are not all equal over the groups with $Y_{jk} > 0$. In particular, for $a_1 < \cdots < a_K$ it suffices that at one death time with $W(t_j) \neq 0$ and $d_j < Y_j$ at least two groups are at risk.
(c) For $K = 2$ and $a_1 < a_2$, show that the trend statistic equals $Z_2(\tau)/\sqrt{\hat{\sigma}_{22}} = -Z_1(\tau)/\sqrt{\hat{\sigma}_{11}}$.

Solution [Supplement].

(a) Each t_j contributes $w_j\{\mathrm{diag}(\boldsymbol{\pi}_j) - \boldsymbol{\pi}_j\boldsymbol{\pi}_j'\}$ to $\hat{\Sigma}$; the brace in the formula is the variance of a_κ for a label $\kappa \sim \boldsymbol{\pi}_j$, hence ≥ 0, and $w_j \geq 0$.

(b) The variance vanishes iff a_k is constant on $\{k : Y_{jk} > 0\}$, and a sum of nonnegative terms is positive iff some term is. For increasing scores, two groups at risk make the a_k unequal, and $W(t_j) \neq 0$, $1 \leq d_j < Y_j$ give $w_j > 0$.

(c) $Z_1 = -Z_2$ and $\hat{\sigma}_{11} = -\hat{\sigma}_{12} = \hat{\sigma}_{22}$ (Exercise 9.3(a)), so

$$\mathbf{a}'\mathbf{Z} = (a_2 - a_1)Z_2, \qquad \mathbf{a}'\hat{\Sigma}\mathbf{a} = (a_2 - a_1)^2\hat{\sigma}_{22}.$$

Divide ($a_2 - a_1 > 0$): the one-sided two-sample test of $h_1 \leq h_2$. ■

Exercise 10.2 (The paired log-rank test as an exact sign test)

In pair i ($i = 1, \ldots, M$, pairs independent) let X_{1i}, X_{2i} be i.i.d. continuous and both censored at a common time C_i independent of (X_{1i}, X_{2i}). Let D_1 (D_2) be the number of pairs in which member 1 (member 2) has an observed event first, and $m = D_1 + D_2$.
(a) Show that, conditionally on m, $D_1 \sim \mathrm{Bin}(m, 1/2)$.
(b) Show that $\mathrm{E}(D_1 - D_2 \mid m) = 0$ and $\mathrm{Var}(D_1 - D_2 \mid m) = m$, matching the paired log-rank statistic.
(c) Compute the exact two-sided p-value for the 6-MP pairs ($D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$).

Solution [Supplement].

(a)

Step 1. Member 1 has an observed event first iff $A_{1i} = \{X_{1i} < X_{2i},\ X_{1i} \leq C_i\}$; likewise A_{2i}; disjoint (ties null).

Step 2. Swapping X_{1i}, X_{2i} preserves the law and maps A_{1i} to A_{2i}: equal probabilities.

Step 3. With $\xi_i = 1, 2, 0$ for A_{1i}, A_{2i}, neither: the ξ_i are independent and $P(\xi_i = 1 \mid \xi_i \neq 0) = 1/2$, so given $J = \{i : \xi_i \neq 0\}$, $D_1 \sim \mathrm{Bin}(|J|, 1/2)$, depending on J only through $m = |J|$.

(b) $D_1 - D_2 = 2D_1 - m$: mean 0, variance m; with $W \equiv 1$ this is the paired statistic $(D_1 - D_2)/\sqrt{D_1 + D_2}$.

(c) With $m = 21$ and $D_1 = 18$,

$$p = 2P\{\mathrm{Bin}(21, \tfrac{1}{2}) \geq 18\} = 2 \cdot \frac{\binom{21}{18} + \binom{21}{19} + \binom{21}{20} + \binom{21}{21}}{2^{21}} = \frac{2(1330 + 210 + 21 + 1)}{2\,097\,152} \approx 0.0015,$$

close to the normal approximation 0.001. ■